

Crosscutting Concepts in Biology

The units of this book seem to cover different subjects, but all biological sciences are tied together by themes and methods of study that cut across disciplines. These concepts overlap and interlock, and crop up throughout the book. You'll also notice that several of these concepts overlap with the characteristics of life or the nature of science.

The study of biology revolves around several crosscutting concepts: cause and effect; systems and system models; stability and change; patterns; scale, proportion, and quantity; energy and matter: flows, cycles, and conservation; and structure and function.

Cause and Effect: Mechanism and Explanation Science is not a list of facts, but "a way of knowing." The job of science is to use observations, questions, and experiments to explain the natural world in terms of natural forces and events, or cause and effect. Successful scientific research reveals rules and patterns that can explain and predict at least some events in nature. Science enables us to take actions that affect events in the world around us.

Systems and System Models Within the biosphere, all of the different parts of the system work together and interact with one another. Predation, competition, and symbiosis within an ecological community are examples of interactions that occur within a biological system. The regulation of blood pressure in the human body can be studied as interactions among the nervous, endocrine, excretory, and circulatory systems.

Stability and Change Living things maintain a relatively stable internal environment, a process known as homeostasis. For most organisms, any breakdown of homeostasis may have serious or even fatal consequences.

Patterns You can compare similar patterns among all sorts of objects in nature, such as those shown in **Figure 1-11**. Patterns can be either linear or cyclical. Linear patterns include the tendency for organ systems to become more complex as you move through vertebrates from fishes to mammals. Cyclical patterns include seasonal migrations of animals or movement of materials such as carbon and nitrogen through the environment.

Scale, Proportion, and Quantity Just about everything, from atoms to rocks to organisms, can be described in terms of structure. How something's structure is studied and described depends on the scale on which it is being studied. On a molecular scale, a scientist might study a human in terms of proteins in the body. On a global scale, a scientist might study humans in terms of their impact on the ozone layer.

Figure 1-11
Patterns in Nature

Throughout nature you will find patterns. Similar patterns can be in structures ranging from an unrolling fern frond to a shell to a pine cone.

